AIAA Journal
Vol. 51, No. 5, May 2013

Technical Notes

Analysis of Numerical Errors Generated by Slowly Moving Shock Waves

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DOI: 10.2514/1.J051884

I. Introduction

Despite being the most common means for simulating shocks, shock capturing has a well known drawback: the generation of numerical errors when shocks move slowly relative to the grid [1,2]. These errors, produced at the shock, propagate downstream and contaminate the solution. This problem is particularly detrimental when the error is on the order of physical fluctuations (e.g., in acoustics and turbulence). Errors due to slowly moving shocks were first identified in supersonic simulations and attributed to vanishing artificial viscosity as $s \to 0$ [3], where s is the shock speed. These errors, of oscillation wavelength $\lambda_s \Delta x / s$, where Δx is the grid spacing and λ_s is the wave speed associated with the oscillations, occur in nonlinear systems [4]. A smeared density profile introduces a momentum spike at the shock [5] for first-order and high-order methods. Intermediate states lie on the nonphysical branch of the Hugoniot locus between upstream and downstream states [6]. Entropy fixes have been developed specifically for this problem [7–9] with varying levels of success. Past studies on shock instabilities suggest that the Harten-Lax-van Leer (HLL) solver may alleviate these problems [10], which may fall into the broader category of carbuncle-related phenomena [11–13]. The present study divides slowly moving shock errors into startup errors produced only initially and unsteady postshock oscillations shed periodically. Both are analyzed and fixes are proposed, for first and high order, before extension to the full two-dimensional (2-D) Euler equations.

II. Preliminaries

For constant temperature, and thus sound speed a, the one-dimensional isothermal Euler equations can be written in semidiscrete form:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\bar{\rho}_i\\ \overline{\rho u}_i\end{pmatrix} = -\frac{1}{\Delta x}\begin{pmatrix}(\rho u)_{i+1/2} - (\rho u)_{i-1/2}\\ (\rho u^2+\rho a^2)_{i+1/2} - (\rho u^2+\rho a^2)_{i-1/2}\end{pmatrix} \quad (1)$$

where u is the velocity, ρ is the density, and the bar is the cell average. The initial focus is on first-order finite-volume discretizations; high-order schemes are discussed in Sec. VI. Third-order total variation diminishing Runge–Kutta time marching is used, which is required for weighted essentially non-oscillatory (WENO) schemes but does not affect the first-order solution. The numerical fluxes can be written as a central term followed by a diffusive component:

$$(\rho u)_{i+1/2} = \frac{(\rho u)_i + (\rho u)_{i+1}}{2} - \{c^{(\rho)}_{i+1/2}(\rho_{i+1}-\rho_i) - c^{(\rho u)}_{i+1/2}[(\rho u)_{i+1} - (\rho u)_i]\} \quad (2)$$

$$(\rho u^2+\rho a^2)_{i+1/2} = \frac{(\rho u^2+\rho a^2)_i + (\rho u^2+\rho a^2)_{i+1}}{2} - \{m^{(\rho)}_{i+1/2}(\rho_{i+1}-\rho_i) - m^{(\rho u)}_{i+1/2}[(\rho u)_{i+1} - (\rho u)_i]\} \quad (3)$$

where c, $m \sim \Delta x$ are components of the numerical viscosity matrix. Different standard Riemann solvers are considered [1]. For the Lax–Friedrichs (LF) and Rusanov Local Lax-Friedrichs (LLF) solvers, the elements of the artificial viscosity matrix are

$$c^{(\rho)}_{i+1/2} = \beta_{i+1/2}/2, \qquad c^{(\rho u)}_{i+1/2} = 0,$$
$$m^{(\rho)}_{i+1/2} = 0, \qquad m^{(\rho u)}_{i+1/2} = \beta_{i+1/2}/2 \quad (4)$$

where $\beta^{\mathrm{LF}}_{i+1/2} = \max_{i=1,N,p=1,2}|\lambda^{(p)}_i|$, N is the number of grid points, and $\beta^{\mathrm{LLF}}_{i+1/2} = \max_{p=1,2}(|\lambda^{(p)}_i|, |\lambda^{(p)}_{i+1}|)$. For the Roe solver, the matrix elements are

$$c^{(\rho)}_{i+1/2} = (|\tilde\lambda^{(1)}_{i+1/2}|\tilde\lambda^{(2)}_{i+1/2} - |\tilde\lambda^{(2)}_{i+1/2}|\tilde\lambda^{(1)}_{i+1/2})/4a,$$
$$c^{(\rho u)}_{i+1/2} = (|\tilde\lambda^{(2)}_{i+1/2}| - |\tilde\lambda^{(1)}_{i+1/2}|)/4a \quad (5)$$

$$m^{(\rho)}_{i+1/2} = -\tilde\lambda^{(1)}_{i+1/2}\tilde\lambda^{(2)}_{i+1/2}c^{(\rho u)}_{i+1/2},$$
$$m^{(\rho u)}_{i+1/2} = c^{(\rho)}_{i+1/2} + (\tilde\lambda^{(1)}_{i+1/2} + \tilde\lambda^{(2)}_{i+1/2})c^{(\rho u)}_{i+1/2} \quad (6)$$

where $\tilde{}$ denotes Roe averages. For the HLL solver, the matrix elements are

$$c^{(\rho)}_{i+1/2} = \frac{-s^-_{i+1/2}s^+_{i+1/2}}{s^+_{i+1/2} - s^-_{i+1/2}}, \qquad c^{(\rho u)}_{i+1/2} = \frac{1}{2}\frac{s^+_{i+1/2} + s^-_{i+1/2}}{s^+_{i+1/2} - s^-_{i+1/2}} \quad (7)$$

$$m^{(\rho)}_{i+1/2} = -\tilde\lambda^{(1)}_{i+1/2}\tilde\lambda^{(2)}_{i+1/2}c^{(\rho u)}_{i+1/2},$$
$$m^{(\rho u)}_{i+1/2} = c^{(\rho)}_{i+1/2} + (\tilde\lambda^{(1)}_{i+1/2} + \tilde\lambda^{(2)}_{i+1/2})c^{(\rho u)}_{i+1/2} \quad (8)$$

where $s^-_{i+1/2} = \min(0, \lambda^{(1)}_i, \tilde\lambda^{(1)}_{i+1/2})$, and $s^+_{i+1/2} = \max(0, \lambda^{(2)}_{i+1}, \tilde\lambda^{(2)}_{i+1/2})$. For isothermal Euler, HLL and Roe are almost identical, with a full matrix of locally varying elements that may be negative, although the matrix is positive–definite. In LF/LLF, the matrix is diagonal with positive components. Initially, the eigenvalues depend on the upstream and downstream velocities, which can be expressed in terms of the shock speed s and Mach number M. Thus, the amount of dissipation is initially prescribed by s and M. For small s, one of the eigenvalues may change sign across the shock.

III. Classification and Characterization of the Errors

Figure 1 shows a time series of an $M = 2$ shock propagating at $s = 0.1$ into a field of density $\rho_R = 1$. In all solvers, an undershoot, or

Presented as Paper 2011-657 at the 49th AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition, Orlando, 4–7 January 2011; received 14 February 2012; revision received 29 November 2012; accepted for publication 2 December 2012; published online 8 March 2013.

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spike [5], in the momentum develops initially and propagates with the shock. Thereafter, the solution depends on the solver. In LF/LLF, the shock adopts a diffuse steady-state profile with a spike. For Roe/HLL, the momentum spike oscillates in an unsteady fashion, shedding waves propagating downstream of the shock; these oscillations are rapidly damped due to the dissipation of first-order schemes. Motivated by past work [5,8], these observations suggest dividing errors into two categories: startup errors and postshock oscillations.

IV. Analysis and Correction of Startup Errors

Startup errors are defined as errors produced at the beginning of the simulation, though they may persist. These errors typically consist of a momentum spike moving with the shock [5] and excess momentum (and mass) propagating downstream of the shock by conservation. These errors apply to all solvers and are most striking in LF/LLF, in which case the shock achieves a steady-state profile that no longer changes with time and moves at s. The traveling-wave analysis of Jin and Liu [5] explains the spike formation by writing

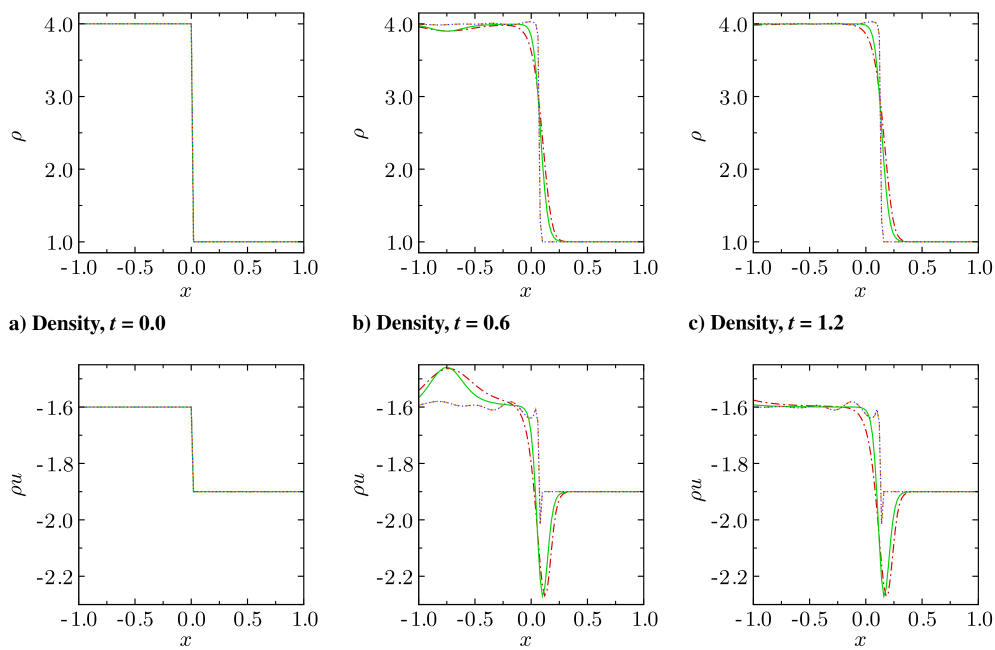
$$\rho u = \rho u_L + s(\rho - \rho_L) + \epsilon \rho' \tag{9}$$

where L denotes the downstream state, the prime differentiation is $\xi = x - st$, and ϵ is the viscosity. Because ρu_L is constant, the prominence of the spike depends on the relative magnitude of $s\rho$ and $\epsilon\rho'$. This analysis is directly applicable to LF with $\epsilon = \beta\Delta x/2 = 0.029$ and $\beta = |\lambda_R^{(1)}| = 2.9$ for $\Delta x = 0.02$. Figure 2 shows simulations results and momentum calculated from Eq. (9) based on the density in the simulations, along with the contribution of each term. The traveling-wave solution matches the momentum from the simulations almost exactly, as expected; this result holds for a broad range of s and Δx. Although ϵ depends on s, the dependence of the last term of Eq. (9) on s is weak, such that the prominence of the spike is mainly governed by the second term, and thus s.

The momentum spike cannot be avoided because of the diffuse shock profile, although it may be removed by postprocessing [5]. On the other hand, the downstream-propagating wave, which may interact with flow features downstream of the shock, can be corrected by initializing the shock profile to the appropriate traveling-wave solution. Once the shock adopts a diffuse profile, density is convected at s and diffuses with artificial viscosity ϵ, such that

$$\rho(x) = \rho_L(1 - \operatorname{erf}\eta)/2 + \rho_R(1 + \operatorname{erf}\eta)/2, \qquad \eta = x/\sqrt{4\epsilon c\Delta x} \tag{10}$$

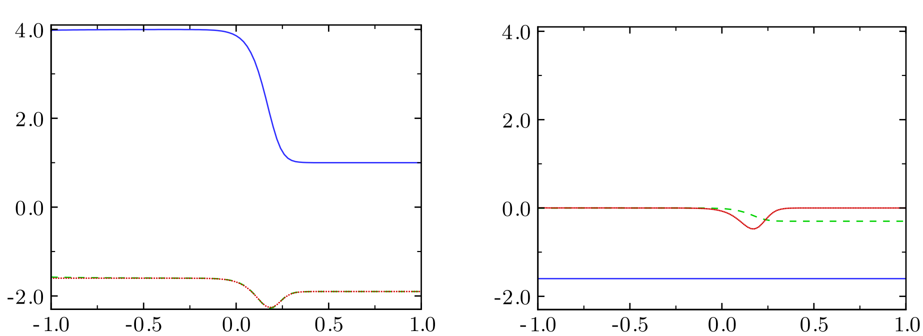
where c is an empirically determined constant that may depend on the scheme but not on Δx; here, $c \approx 5$ yields a very small error.



a) Density, $t = 0.0$ b) Density, $t = 0.6$ c) Density, $t = 1.2$

d) Momentum, $t = 0.0$ e) Momentum, $t = 0.6$ f) Momentum, $t = 1.2$

Fig. 1 Shock evolution for different solvers ($M = 2$, $s = 0.1$, $\Delta x = 0.02$): LF (red dash-dotted), LLF (green solid), Roe (orange dashed), and HLL (blue dotted, indistinguishable from Roe).



a) Density (blue solid), momentum (green dashed) and traveling wave solution (red dotted)

b) ρu_L (blue solid), $s(\rho - \rho_L)$ (green dashed), and $\epsilon\rho'$ (red dotted) in Eq. (9)

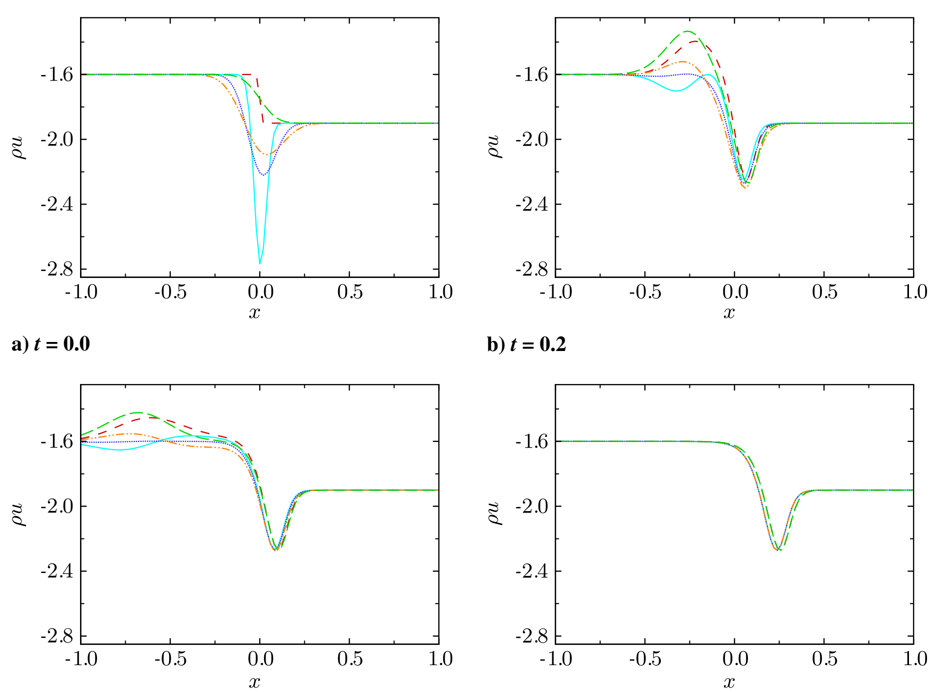
Fig. 2 Agreement between the traveling wave and computed solutions (LF, $t = 1.2$, $M = 2$, $s = 0.1$, $\Delta x = 0.02$).

Momentum can then be computed from Eq. (9). It follows that arbitrary smearing of the initial shock profile [14] leads to the generation of a downstream-propagating wave, as shown in Fig. 3, where different initializations are used. Regardless of the smearing, the same steady-state solution is achieved.

V. Analysis and Correction of Postshock Oscillations

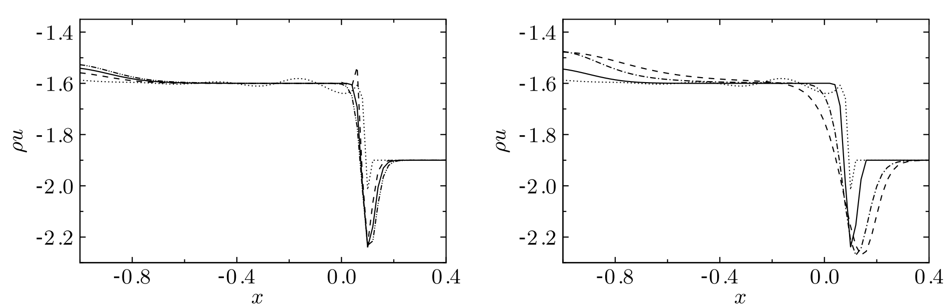
When using Roe/HLL, unsteady postshock oscillations are generated as the shock crosses grid points. These oscillations do not occur in first-order LF/LLF; Roe/HLL solvers are different in two main aspects: elements of the artificial viscosity matrix may be negative, and the numerical diffusion terms include contributions from both ρ and ρu. When the shock propagates into a new cell, a momentum spike grows at the shock; by conservation, a wave of positive amplitude propagating downstream is produced. When considering HLL, $c^{(\rho)}$ is zero, while the $c^{(\rho u)}$ is negative in the artificial viscosity matrix. Because $s^+ = 0$ ahead of the shock, $c^{(\rho)} = 0$ and $c^{(\rho u)} = -1/2$, although $m^{(\rho)}, m^{(\rho u)} > 0$ at that point just ahead of the shock. When the shock enters that cell, the diffusive term in the continuity equation has a negative viscosity and thus has a destabilizing effect. As the shock fills the cell, the eigenvalue of the shock family, and therefore $c^{(\rho)}$, become positive in that cell, thus now leading to a diffusive and stabilizing behavior. This phenomenon is confirmed by examination of the solution. The largest spike occurs when the local velocity in the cell is $u = -1$; thereafter, the eigenvalue and $c^{(\rho)}$ remain positive. Once the shock propagates into the next cell, the process repeats itself, thus leading to the periodic shedding. The frequency of postshock oscillations increases with grid refinement, but the amplitude does not decrease. The spike generated in the first cell is the largest and is attributed to the startup error. Although the traveling-wave analysis cannot be applied directly, an approximate approach confirms spike formation. The momentum satisfies $-c^{(\rho u)}(\rho u)' + \rho u = \rho u_L + s(\rho - \rho_L) + c^{(\rho)}\rho'$. Assuming that the density is sufficiently smooth and by "freezing" the solution in time, the coefficients can be determined based on the velocity. Following the same development as in Sec. IV, the solution yields a spike.

These arguments suggest that postshock oscillations can be prevented by imposing a lower bound on s^+ when $\lambda^{(2)}$ changes sign, thus ensuring sufficient dissipation. Figure 4 shows momentum profiles for different lower bounds on s^+. Although decreasing s^+ reduces the dissipation, postshock oscillations occur beyond a certain value. The critical s^+ that exhibits the least dissipation while



a) $t = 0.0$ b) $t = 0.2$ c) $t = 0.5$ d) $t = 2.0$

Fig. 3 Momentum profiles for different initializations [Eq. (10), LF, $M = 2$, $s = 0.1$, $\Delta x = 0.02$]: one-point jump (red dashed), smeared momentum (green long-dashed), $c = 1$ (cyan solid), $c = 5$ (blue dotted), $c = 10$ (orange dashed-dot-dot).



a) $s^+ = 0.1$ (dotted), 0.4 (dashed), 0.6 (solid), 0.8 (dash-dotted)

b) Roe (dotted), LF (dashed), LLF (dash-dotted), HLL-fix (solid)

Fig. 4 Momentum profiles ($M = 2$, $s = 0.1$, $\Delta x = 0.02$) to illustrate the performance of the proposed fix.

preventing oscillations is $s^+ = 0.6$, which corresponds to $\lambda_L^{(2)}$ for $M = 2$ and $s = 0.1$. Following extensive testing, setting the lower bound to the eigenvalue associated with the shock family based on the downstream properties leads to optimal behavior for a wide range of s and M. Then, the momentum spike becomes steady and propagates with the shock, similar to LF but less dissipative. A localized version can be implemented, i.e., applied only near slowly moving shocks defined by a sign change in the corresponding eigenvalue, $\lambda_i^{(2)}\lambda_{i+1}^{(2)} < 0$ for $\partial\rho/\partial x < 0$ across the shock, and expanding the region by two points on either side for robustness. Bounding the wave speed in $c^{(\rho)}$ is sufficient; other fields need not be affected. In summary, the algorithm to prevent postshock oscillations is

$$\text{if } \lambda_i^{(2)}\lambda_{i+1}^{(2)} < 0, \quad \forall\, j \in [i-2, i+2],$$

$$c_{j+1/2}^{(\rho)} = \frac{-s_{j+1/2}^{-}s_{j+1/2}^{+,\text{mod}}}{s_{j+1/2}^{+,\text{mod}} - s_{j+1/2}^{-}}, \quad \text{where } s^{+,\text{mod}} = \max(s^+, |\lambda_L^{(2)}|) \tag{11}$$

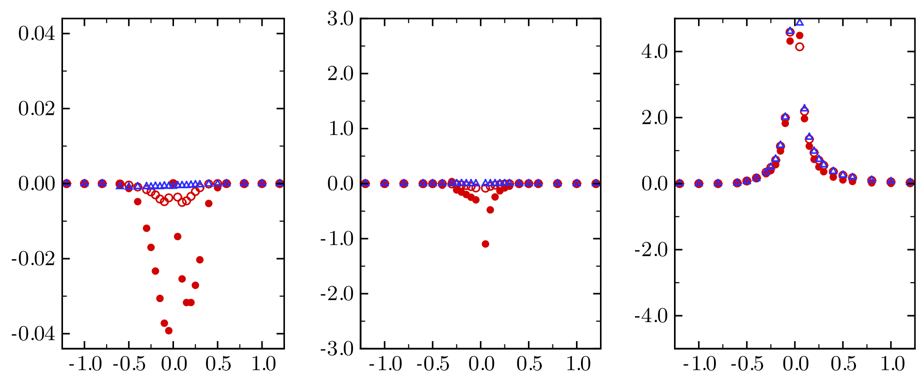
For $\partial\rho/\partial x > 0$, the downstream state is the right state, and the fix is applied where $\lambda_i^{(1)}\lambda_{i+1}^{(1)} < 0$. The performance of the present entropy fix is assessed by computing the error after the initial transient, $\Delta f = (f_{\text{exact}} - f_{\text{computed,max}})/|f_L - f_R|$. A positive/negative value implies undershoot/overshoot. Figure 5 shows the overshoot in density, the overshoot in momentum downstream of the shock, and the undershoot of the spike versus shock speed for $M = 4$ with Roe, Roe with Karni and Canic's entropy fix [8], and the present HLL fix [Eq. (11)]. Errors with the Roe fix are small, but with the HLL fix, they are effectively negligible. The error due to the spike can be large (up to 300%) for all cases.

VI. High-Order Accurate Schemes

Higher-order schemes, such as WENO [15] exhibit postshock oscillations [5], even with LF and the HLL fix, in contradiction with first-order results. The problem is more serious because of low dissipation away from shocks. As illustrated in Fig. 6a, the error amplitude increases and its wavelength decreases with order of accuracy. The oscillations are explained by writing the flux difference for LF as a central term followed by a diffusive component:

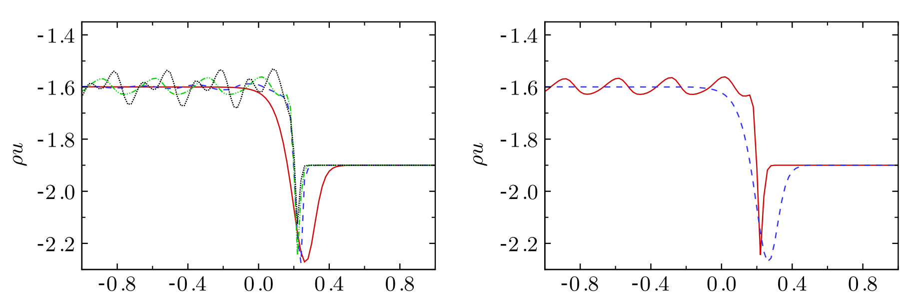
$$\frac{f_{i+1/2} - f_{i-1/2}}{\Delta x} = \frac{\frac{f(q_{i+1/2}^R)+f(q_{i+1/2}^L)}{2} - \frac{f(q_{i-1/2}^R)+f(q_{i-1/2}^L)}{2}}{\Delta x} - \frac{\beta^{\text{LF}}}{2}\frac{\frac{(q^R-q^L)\Delta z}{\Delta z}\big|_{i+1/2} - \frac{(q^R-q^L)\Delta z}{\Delta z}\big|_{i-1/2}}{\Delta x} \tag{12}$$

where $q_{i+1/2}^{L/R}$ are the left/right reconstructed states in finite-volume WENO, $\Delta z = (q^R - q^L)/\frac{\partial q}{\partial x}$, and $\frac{\partial q}{\partial x}$ is locally approximated. The diffusive term is a low-order approximation of the high-order first derivative. Increasing the order of accuracy for fixed N reduces the difference between $q_{i+1/2}^R$ and $q_{i+1/2}^L$, and thus Δz, for a sufficiently smooth solution. Thus, the artificial viscosity $\beta^{\text{LF}}\Delta z/2$ decreases with increasing order of accuracy, preventing sufficient dissipation near shocks and leading to overshoots/undershoots. To correct these errors, the diffusive component of the numerical flux $f_{i+1/2}$ is written as the following pure diffusion term $\beta^{\text{LF}}(q_{i+1} - q_i)/2$, instead of the usual $\beta^{\text{LF}}(q_{i+1/2}^R - q_{i+1/2}^L)/2$ form; a high-order expression is readily implemented. No special treatment is required for multiple dimensions. As shown in Fig. 6b, this definition prevents oscillations, although with more dissipation at the shock. This scheme needs to be applied only near slowly moving shocks (as described in Sec. V). The



a) Density overshoot b) Momentum overshoot c) Momentum spike

Fig. 5 Normalized density and momentum errors vs. s for different solvers ($M = 4, \Delta x = 0.02$): Roe (red filled circles), Roe fix [8] (red open circles), and HLL fix (blue open triangles).



a) WENO1 (red solid) WENO3 (blue dashed); WENO5 (green dashed-dot-dot); WENO7 (black dotted)

b) WENO5 using the standard (red solid) and modified flux from Eq. (31) (blue dashed)

Fig. 6 Momentum profiles to show the occurrence and prevention of postshock oscillations for high-order WENO with LF ($M = 2, s = 0.1, \Delta x = 0.02$).

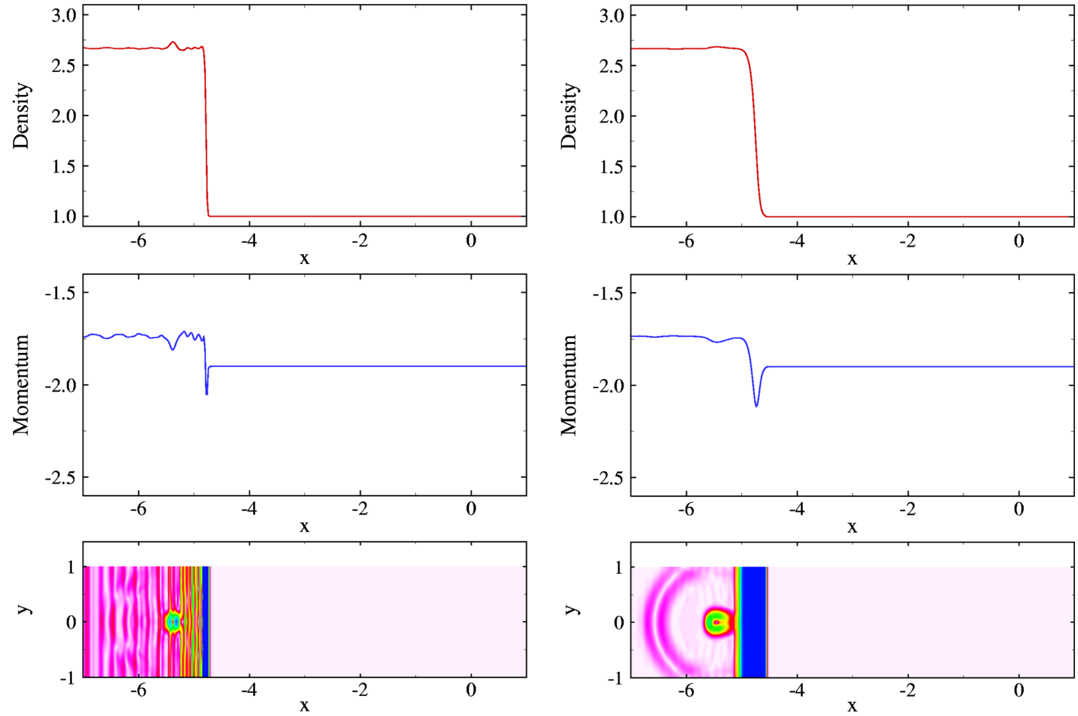


Fig. 7 2-D shock–entropy-wave interaction (WENO5, $s = 0.1$, $M = 2$, $\Delta x = 0.02$). Top/middle: density/momentum along the centerline; bottom: numerical Schlieren contours.

transition between the base scheme and the fix can be smoothed, but the required number of points may be considerable. For WENO5, the oscillation amplitude in the momentum is 15% with no fix; the amplitude reduces to 5% with two points upstream and downstream and to 0.02% with 10 points downstream and five upstream. Similar to a local LF approach, setting β^{LF} to $|\lambda_L^{(2)}|$ further reduces dissipation.

VII. Full Euler Equations in Multiple Dimensions

Extension to the full multi-dimensional Euler equations is trivial. The 2-D interaction of a slowly moving shock with an entropy wave is shown in Fig. 7 for Roe's scheme and LF fix using WENO5. If no fix is used, postshock oscillations die out so slowly that the physical waves generated upon interaction are barely discernible. The proposed scheme does not exhibit such oscillations, although at the expense of additional dissipation at the shock. A possibly more-efficient implementation would be using first order near slowly moving shocks only and high order elsewhere. Similar errors have been reported in the 2-D shock/vorticity wave interaction [16]; although results are not shown here for conciseness, postshock oscillations can be shown to be prevented only if the fix is applied in both x and y because of the two-dimensionality of the incoming perturbation.

VIII. Conclusions

Errors generated by slowly moving shocks are analyzed and divided into two categories: startup errors observed with all solvers and postshock oscillations produced by the Roe and Harten-Lax-van Leer (HLL) solvers. It is shown that, by smearing the initial profiles appropriately for Lax–Friedrichs (LF) and Rusanov solvers, the downstream-propagating wave in the startup error is prevented. Postshock oscillations are removed by setting a lower bound on the HLL wave speed. For high order, even LF leads to postshock oscillations; an additional fix is proposed, in which the flux difference is written in terms of central and numerical diffusion terms. The fixes are local and have been tested on several problems, including two-dimensional shock-entropy wave interaction with the full Euler equations.

Acknowledgments

This work was primarily performed at the Center for Turbulence Research (Stanford University) and was supported by Department of Energy grant DE-FC02-06-ER25787. The author thanks S. Lele, J. Larsson, S. Kawai, D. Zaide, and P. Roe for useful discussions.

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Associate Editor